

Data Mining Methods: Final Exam Comprehensive Study Guide

This guide is designed to prepare you for a 40-question multiple-choice final exam by consolidating definitions, core concepts, and a bank of 50 practice questions (bringing your total practice pool to over 80 questions when combined with our previous discussion).

Part 1: Key Definitions & Core Concepts

General Foundations

- **Data Mining Pipeline:** A multi-step process including data understanding, pre-processing, warehousing, data modeling, and evaluation.
- **Knowledge Discovery Views:** Projects are viewed through data types, application domains, discovered knowledge, and techniques.
- **Supervised vs. Unsupervised Learning:** Supervised learning (Classification) uses predefined class labels as ground truth. Unsupervised learning (Clustering) identifies natural groupings without predefined labels.
- **Curse of Dimensionality:** As dimensions increase, data becomes sparse and distance-based separation becomes fuzzy, making traditional algorithms ineffective.

Classification (Supervised)

- **Decision Tree Induction:** A top-down, "divide and conquer" greedy approach that uses attributes to split data into subsets. It relies on **Entropy** (randomness) and **Information Gain** (reduction in entropy) to select the best split.
- **Naive Bayes:** A probabilistic method based on class-conditional probabilities. It is **incremental**, meaning it can be updated with new data without retraining from scratch.
- **Support Vector Machines (SVM):** Finds the optimal hyperplane with the maximum margin to separate classes.
- **Neural Networks:** Mimics human neurons using layers (input, hidden, output). **Backpropagation** adjusts link weights from right-to-left based on the error at the output layer.
- **Ensemble Methods:**
 - **Bagging:** Uses multiple models with **equal weights** and majority voting.
 - **Boosting:** Assigns **different weights** to models, prioritizing cases previously misclassified.

Clustering (Unsupervised)

- **K-means:** A partitioning method that assigns objects to the nearest **centroid** (mean value) and updates centroids iteratively. Time complexity is $O(nkt)$.
- **K-medoids:** Uses an **actual object (medoid)** as a representative center instead of a mean value, making it more robust to outliers.
- **Hierarchical Clustering:** Creates a tree-like structure (**Dendrogram**). It can be **Agglomerative** (bottom-up merging) or **Divisive** (top-down splitting).
- **DBSCAN:** A density-based method that defines clusters as dense regions separated by sparse areas. It uses **Epsilon** (radius) and **MinPoints** (density threshold).
- **EM (Expectation-Maximization):** A probabilistic approach using **Gaussian Mixture Models**.
 - **E-step:** Calculates the probability (responsibility) of an object belonging to a cluster.
 - **M-step:** Updates cluster parameters (mean, standard deviation) based on those probabilities.

Anomaly Detection (Outlier Analysis)

- **Global Outliers:** Deviate from the entire dataset.
- **Contextual Outliers:** Abnormal only within a specific context (e.g., temperature spikes in winter).
- **Collective Outliers:** A group of objects that deviate together from the norm.
- **Proximity-Based Detection:** Assumes outliers are far from others. This can be measured via **distance** (absolute proximity) or **density** (relative proximity).

Part 2: Multiple Choice Question Bank

1. Which view of data mining characterizes a project by what the attributes and characteristics are?

- A. Technique View
- B. Data View
- C. Knowledge View
- D. Application View

Answer: B. Data View

2. Which stage of the data mining pipeline is focused on managing multi-dimensional data analysis?

- A. Pre-processing
- B. Data Understanding
- C. Data Warehousing
- D. Evaluation

Answer: C. Data Warehousing

3. In a Decision Tree, what does "Entropy" measure?

- A. The number of nodes in a tree
- B. The degree of randomness or impurity in a dataset
- C. The accuracy of the classification
- D. The speed of the algorithm

Answer: B. The degree of randomness or impurity in a dataset

4. Which classification method is specifically noted for its solid probabilistic foundation and incremental nature?

- A. SVM
- B. Decision Trees
- C. Bayesian Classification
- D. Neural Networks

Answer: C. Bayesian Classification

5. What occurs in a Neural Network during "Backpropagation"?

- A. Data flows from the input layer to the output layer.
- B. Centroids are updated based on mean values.
- C. Weights are adjusted from the output layer back toward the input layer based on error.
- D. The number of hidden layers is doubled.

Answer: C. Weights are adjusted from the output layer back toward the input layer based on error.

6. Which ensemble method uses majority voting and treats all models as having equal weight?

- A. Bagging
- B. Boosting
- C. Backpropagation
- D. K-means

Answer: A. Bagging

7. In a Confusion Matrix, "Sensitivity" is the ratio of:

- A. True Positives to all actual positives
- B. True Positives to all objects predicted as positive
- C. True Negatives to all predicted negatives
- D. Correct predictions to the total number of objects

Answer: A. True Positives to all actual positives

8. What is "Precision" in a binary classification evaluation?

- A. $TP / (TP + FN)$
- B. $TP / (TP + FP)$
- C. $TN / (TN + FP)$
- D. $(TP + TN) / \text{Total}$

Answer: B. $TP / (TP + FP)$

9. A "Random Classifier" on an ROC curve is represented by:

- A. A curve in the top-left corner
- B. A diagonal line from (0,0) to (1,1)
- C. A vertical line at $x=0.5$
- D. A horizontal line at $y=1.0$

Answer: B. A diagonal line from (0,0) to (1,1)

10. Which statistical test determines if the difference in error between two models is significant or by chance?

- A. ROC Curve
- B. T-test
- C. Confusion Matrix
- D. Silhouette Coefficient

Answer: B. T-test

11. What is the time complexity of the K-means algorithm?

- A. $O(n)$
- B. $O(\log n)$
- C. $O(nkt)$
- D. $O(n \times m)$

Answer: C. $O(nkt)$

12. Which clustering method is more robust to outliers because it uses an actual object as a center?

- A. K-means
- B. K-medoids
- C. DBSCAN
- D. EM Clustering

Answer: B. K-medoids

13. A "Dendrogram" is a visual tool used for which type of clustering?

- A. Partitioning
- B. Hierarchical
- C. Grid-based
- D. Density-based

Answer: B. Hierarchical

14. In DBSCAN, an object that is reachable from a core object but is not a core object itself is a:

- A. Noise object
- B. Centroid
- C. Border object
- D. Medoid

Answer: C. Border object

15. In EM Clustering, what is calculated during the "E-step"?

- A. The probability of each point belonging to each cluster
- B. The new mean of the Gaussian distribution
- C. The updated standard deviation
- D. The number of clusters K

Answer: A. The probability of each point belonging to each cluster

16. Which method is used to find clusters in a subset of dimensions for high-dimensional data?

- A. Bi-clustering
- B. Subspace Clustering
- C. Partitioning
- D. Density-based Clustering

Answer: B. Subspace Clustering

17. What is the goal of "Bi-clustering"?

- A. Dividing data into exactly two clusters
- B. Comparing two different clustering algorithms
- C. Jointly grouping both objects and dimensions
- D. Using two different distance functions

Answer: C. Jointly grouping both objects and dimensions

18. Community Detection in social networks is often captured through:

- A. K-means
- B. Minimal cuts in a graph
- C. Backpropagation
- D. Entropy calculation

Answer: B. Minimal cuts in a graph

19. Which type of outlier analysis requires context (like time or location) to identify anomalies?

- A. Global Outlier
- B. Contextual Outlier
- C. Collective Outlier
- D. Structural Outlier

Answer: B. Contextual Outlier

20. A "Super Object" is used to transform which problem into a global outlier detection problem?

- A. Contextual Anomaly
- B. Collective Anomaly
- C. Sequential Pattern Mining
- D. Decision Tree Splits

Answer: B. Collective Anomaly

21. In Time Series analysis, "Overall Trend" refers to:

- A. The long-term direction of the data (up, down, or flat)
- B. Seasonal fluctuations like annual changes
- C. Random noise in the signal
- D. Anomalous spikes in the data

Answer: A. The long-term direction of the data (up, down, or flat)

22. Which graph mining task predicts if a connection will form between two nodes in the future?

- A. Community Detection
- B. Frequent Subgraph Mining
- C. Link Prediction
- D. Subspace Clustering

Answer: C. Link Prediction

23. "Cyber Safety" in data mining is primarily focused on:

- A. Making websites load faster
- B. Improving the visual resolution of social media images
- C. Detecting malicious accounts and harmful behaviors at scale
- D. Predicting the stock price of social media companies

Answer: C. Detecting malicious accounts and harmful behaviors at scale

24. What is "Data Fusion"?

- A. Deleting duplicate records from a database
- B. Merging different types of data (text, image, etc.) to find hidden relationships
- C. Converting numerical data into categorical labels
- D. The process of splitting a training set and a testing set

Answer: B. Merging different types of data (text, image, etc.) to find hidden relationships

25. A "Knowledge Graph" is a research frontier focused on:

- A. Integrating vast volumes of online information into a graph representation
- B. Creating a flowchart of a company's hierarchy
- C. Mapping the physical locations of servers
- D. Visualizing the error rate of a Neural Network

Answer: A. Integrating vast volumes of online information into a graph representation

26. Which type of learning is used when only a few objects have labels and the rest are unlabeled?

- A. Supervised Learning
- B. Unsupervised Learning
- C. Semi-supervised Learning
- D. Transfer Learning

Answer: C. Semi-supervised Learning

27. "Transfer Learning" involves:

- A. Moving data from a warehouse to a model
- B. Taking knowledge learned in one domain and applying it to another
- C. Transferring weights from the output layer to the input layer
- D. Shifting clusters from K-means to DBSCAN

Answer: B. Taking knowledge learned in one domain and applying it to another

28. Which evaluation metric is often becoming more important than simple accuracy in real-world industry AI?

- A. Information Gain
- B. Interpretability and Explainability
- C. The number of iterations T
- D. Initial centroid location

Answer: B. Interpretability and Explainability

29. What is the purpose of "Pooling" in a Convolutional Neural Network (CNN)?

- A. To add more training data
- B. To aggregate information and reduce dimensionality
- C. To calculate the Information Gain of a feature
- D. To define the Epsilon neighborhood

Answer: B. To aggregate information and reduce dimensionality

30. In the KDD community, the "Applied Data Science Track" focuses on:

- A. Theoretical mathematical proofs
- B. Practical challenges and solutions in industry
- C. Only biological sequence alignment
- D. Developing new programming languages

Answer: B. Practical challenges and solutions in industry

31. Which of the following is NOT a core functionality of data modeling mentioned in the sources?

- A. Classification
- B. Clustering
- C. Outlier Analysis
- D. Data Storage

Answer: D. Data Storage

32. "Intracluster Similarity" should be ____ and "Intercluster Similarity" should be ____?

- A. High; Low
- B. Low; High
- C. High; High
- D. Low; Low

Answer: A. High; Low

33. Which algorithm would be best to find a cluster shaped like a crescent moon?

- A. K-means
- B. DBSCAN
- C. EM Clustering
- D. K-medoids

Answer: B. DBSCAN

34. What does the "M" in EM Clustering stand for?

- A. Medoid
- B. Minimum
- C. Maximization
- D. Mean

Answer: C. Maximization

35. If you have 1,000 normal transactions and only 2 fraud transactions, you have a:

- A. Curse of Dimensionality
- B. Class Imbalance Problem
- C. Clustering Tendency
- D. Frequent Itemset

Answer: B. Class Imbalance Problem

36. Which approach uses "Gaussian Mixture Models"?

- A. Naive Bayes
- B. Probabilistic Clustering (EM)
- C. Decision Trees
- D. Link Prediction

Answer: B. Probabilistic Clustering (EM)

37. In proximity-based outlier detection, "Relative Proximity" is associated with:

- A. Distance-based methods
- B. Density-based methods
- C. K-means centroids
- D. Decision Tree splitting

Answer: B. Density-based methods

38. "Bagging" is an ensemble technique also known as:

- A. Bootstrap Aggregating
- B. Backpropagation
- C. Bayesian Updating
- D. Bi-clustering

Answer: A. Bootstrap Aggregating

39. If you want to see how a model's performance changes over time, you use:

- A. Confusion Matrix
- B. Trend and Evolution Analysis
- C. Information Gain
- D. Epsilon radius

Answer: B. Trend and Evolution Analysis

40. The V value (Degree of Freedom) in a 10-fold cross-validation T-test is usually:

- A. 10
- B. 9
- C. 1
- D. 0

Answer: B. 9

41. Which sequence alignment method breaks a sequence into small snippets to find matches?

- A. K-means
- B. BLAST
- C. DBSCAN
- D. SVM

Answer: B. BLAST

42. Which data type is described as an "ordered list of items"?

- A. Graph Data
- B. Sequence Data
- C. Spatial Data
- D. Relational Data

Answer: B. Sequence Data

43. A "Knowledge Graph" is primarily an example of:

- A. A simple spreadsheet
- B. Data Fusion on the web
- C. A decision tree
- D. A single-layer neural network

Answer: B. Data Fusion on the web

44. "Explainable AI" is a research effort to solve the "black box" problem of:

- A. Naive Bayes
- B. Decision Trees
- C. Neural Networks
- D. K-means

Answer: C. Neural Networks

45. Which clustering method divides the object space into cells?

- A. Hierarchical
- B. Partitioning
- C. Grid-based
- D. Density-based

Answer: C. Grid-based

46. In the T-test, if your calculated T-value is HIGHER than the critical value, you:

- A. Reject the null hypothesis (models are significantly different)
- B. Accept the null hypothesis (models are the same)
- C. Restart the clustering process
- D. Add more dimensions to the data

Answer: A. Reject the null hypothesis (models are significantly different)

47. "Support" and "Confidence" are terms associated with:

- A. Outlier Analysis
- B. Association Rules
- C. Backpropagation
- D. Medoid selection

Answer: B. Association Rules

48. Which outlier type is a spike that is "way above the rest of the dataset"?

- A. Global Outlier
- B. Contextual Outlier
- C. Collective Outlier
- D. Normal Noise

Answer: A. Global Outlier

49. Which of the following is NOT an advantage of Decision Trees?

- A. Efficiency
- B. Interpretability
- C. Incremental processing
- D. Easy to visualize

Answer: C. Incremental processing

50. The "Responsibility" of a point in EM clustering refers to its:

- A. Physical distance from the center
- B. Probabilistic membership to a cluster
- C. Weight in a Boosting ensemble
- D. Information Gain value

Answer: B. Probabilistic membership to a cluster

Additional Practice Questions (51–70)

51. What is the main objective of the "E-step" in the Expectation-Maximization algorithm?

- A. To update the mean and standard deviation of a cluster.
- B. To calculate the probability of an object belonging to each cluster.
- C. To randomly select initial centroids.
- D. To remove outliers from the dataset.

Answer: B. To calculate the probability of an object belonging to each cluster.

52. In a decision tree, which value should be minimized to find the most "pure" split?

- A. Entropy
- B. Information Gain
- C. Number of leaf nodes
- D. Precision

Answer: A. Entropy

53. Which term refers to the process of finding patterns that occur in a specific order?

- A. Frequent Itemset Mining
- B. Sequential Pattern Mining
- C. Clustering
- D. Anomaly Detection

Answer: B. Sequential Pattern Mining

54. Support Vector Machines (SVM) aim to find a hyperplane that:

- A. Connects all data points in a single line.
- B. Separates classes with the maximum possible margin.
- C. Minimizes the number of attributes used.
- D. Replaces centroids in K-means.

Answer: B. Separates classes with the maximum possible margin.

55. In the context of "Cyber Safety," what is the primary role of data mining?

- A. To increase user engagement on social media.
- B. To detect malicious accounts and harmful behaviors at scale.
- C. To suggest new friends to users.
- D. To optimize the resolution of uploaded videos.

Answer: B. To detect malicious accounts and harmful behaviors at scale.

56. Which clustering method divides the data space into a multi-resolution structure?

- A. Density-based
- B. Partitioning
- C. Grid-based
- D. Hierarchical

Answer: C. Grid-based

57. A "Contextual Outlier" is an object that is:

- A. Abnormal relative to the entire dataset regardless of circumstances.
- B. Abnormal only within a specific environment or condition (e.g., temperature in winter).
- C. Part of a group of objects that are collectively abnormal.
- D. A result of a data entry error.

Answer: B. Abnormal only within a specific environment or condition (e.g., temperature in winter).

58. What does "Bagging" stand for in ensemble learning?

- A. Bootstrap Aggregating
- B. Bayesian Aggregating
- C. Backpropagation Grouping
- D. Binary Grouping

Answer: A. Bootstrap Aggregating

59. In the T-test for model selection, if the calculated T-value is greater than the critical value:

- A. You reject the null hypothesis (there is a significant difference).
- B. You accept the null hypothesis (there is no significant difference).
- C. You must restart the 10-fold cross-validation.
- D. The model accuracy is guaranteed to be 100%.

Answer: A. You reject the null hypothesis (there is a significant difference).

60. Which technique is used to jointly group both objects and dimensions?

- A. Subspace Clustering
- B. Bi-clustering
- C. K-medoids
- D. Naive Bayes

Answer: B. Bi-clustering

61. What is the time complexity of the K-means algorithm?

- A. $O(n^2)$
- B. $O(nkt)$
- C. $O(\log n)$
- D. $O(n)$

Answer: B. $O(nkt)$

62. Which metric is the ratio of True Positives to all actual positive cases?

- A. Sensitivity (Recall)
- B. Precision
- C. Specificity
- D. Accuracy

Answer: A. Sensitivity (Recall)

63. "Transfer Learning" is best described as:

- A. Moving data from one warehouse to another.
- B. Applying knowledge learned from one domain to a different but related domain.
- C. Transferring weights from the input layer to the hidden layer.
- D. Converting a decision tree into a set of rules.

Answer: B. Applying knowledge learned from one domain to a different but related domain.

64. Which layer in a Neural Network is responsible for the final classification decision?

- A. Input Layer
- B. Hidden Layer
- C. Output Layer
- D. Pooling Layer

Answer: C. Output Layer

65. What is "Data Fusion"?

- A. Deleting redundant data from a database.
- B. Merging different types of data (text, images, etc.) to find hidden relationships.
- C. The process of splitting data into 80/20 for training and testing.
- D. Combining two clusters into one in hierarchical clustering.

Answer: B. Merging different types of data (text, images, etc.) to find hidden relationships.

66. In DBSCAN, what is a "Core Object"?

- A. An object with a neighborhood density above a certain threshold (MinPoints).
- B. The mean value of all points in a cluster.
- C. An object at the very edge of a cluster.
- D. A point that is considered noise.

Answer: A. An object with a neighborhood density above a certain threshold (MinPoints).

67. Which graph mining task predicts future connections between nodes?

- A. Community Detection
- B. Link Prediction
- C. Frequent Subgraph Mining
- D. Minimal Cut Analysis

Answer: B. Link Prediction

68. Which error metric is most suitable for stock price prediction where the scale of the price matters relative to the error?

- A. Mean Absolute Error
- B. Relative Squared Error
- C. Confusion Matrix
- D. T-test

Answer: B. Relative Squared Error

69. What is a "Dendrogram"?

- A. A type of neural network.
- B. A tree-like diagram showing the sequence of merges or splits in hierarchical clustering.
- C. A plot of the True Positive Rate vs. False Positive Rate.
- D. A grid of cells used in subspace clustering.

Answer: B. A tree-like diagram showing the sequence of merges or splits in hierarchical clustering.

70. Which frontier of data mining focuses on integrating online information into a graph of interconnected entities?

- A. Social Network Analysis
- B. Knowledge Graph
- C. Cyber Safety
- D. Data Warehousing

Answer: B. Knowledge Graph

Part 2: Methods Final Practice Question Bank

1. "Bayesian classification is not an incremental process, meaning it must be completely rebuilt when new data is added." — Is this statement true or false?

- A. True
- B. False

Answer: B. False

2. Which of the following refers to the degree of randomness or impurity in a dataset used for attribute splitting?

- A. Information Gain
- B. Entropy
- C. Accuracy
- D. Support

Answer: B. Entropy

3. For high-quality clustering results, which of the following properties should be true? A. Intracluster similarity should be low. B. Intercluster similarity should be high. C. Intracluster similarity should be high. D. Intercluster similarity should be equal to intracluster similarity.

Answer: C. Intracluster similarity should be high

4. "Decision tree induction is a top-down, greedy 'divide and conquer' approach to classification." — Is this statement true or false?

- A. True
- B. False

Answer: A. True

5. What is the time complexity for performing K-means clustering for T iterations with N objects and K clusters?

- A. $O(n)$
- B. $O(n^2)$
- C. $O(nkt)$
- D. $O(\log n)$

Answer: C. $O(nkt)$

6. Which of the following is not a recognized type of outlier in anomaly detection?

- A. Global outlier
- B. Contextual outlier
- C. Collective outlier
- D. Relational outlier

Answer: D. Relational outlier

7. Support Vector Machines (SVM) aim to find the optimal hyperplane to separate classes with the:

- A. Smallest margin
- B. Maximum margin
- C. Least number of dimensions
- D. Highest entropy

Answer: B. Maximum margin

8. "In EM clustering, the E-step (Expectation step) is used to update the mean and standard deviation of the Gaussian mixture models." — Is this statement true or false?

- A. True
- B. False

Answer: B. False (This occurs during the M-step; the E-step computes probability/responsibility)

9. Which of the following is not a type of ensemble method used to combine multiple models?

- A. Bagging
- B. Boosting
- C. Bootstrapping
- D. Backpropagation

Answer: D. Backpropagation (This is an optimization method for individual neural networks, not an ensemble strategy)

10. Which visual tool represents the multi-level tree structure of merges or splits in hierarchical clustering?

- A. Scatter plot
- B. Boxplot
- C. Dendrogram
- D. ROC Curve

Answer: C. Dendrogram

11. "In frequent pattern analysis, the discovery of a strong correlation between two items implies a direct causal relationship." — Is this statement true or false?

- A. True
- B. False

Answer: B. False

12. Which parameters are used to define clusters as dense regions separated by sparse areas in the DBSCAN algorithm?

- A. K and T
- B. Mean and standard deviation
- C. Epsilon (ϵ) and MinPoints
- D. Support and Confidence

Answer: C. Epsilon (ϵ) and MinPoints

13. Which of the following is not considered an advanced data mining method for complex data types?

- A. Sequence data mining
- B. Graph mining
- C. Web mining
- D. Data Warehousing

Answer: D. Data Warehousing (This is a component of the pipeline, not a modeling method)

14. Which method is specifically used in biological sequence alignment to break a sequence into small snippets to find matches?

- A. BLAST
- B. DBSCAN
- C. Naive Bayes
- D. SVM

Answer: A. BLAST

15. "In a neural network, backpropagation is the process of adjusting weights from the output layer back toward the input layer." — Is this statement true or false?

- A. True
- B. False

Answer: A. True

16. Which of the following is not a key characteristic of K-medoids when compared to K-means?

- A. It uses an actual object as a representative center.
- B. It is more robust to noise and outliers.
- C. It is more computationally expensive.
- D. It always produces identical results regardless of initial medoid selection.

Answer: D. It always produces identical results regardless of initial medoid selection. (Initial selection still impacts the final local optimum)

17. The detection of malicious accounts and harmful behaviors on Online Social Networks (OSNs) is a core focus of:

- A. Data Warehousing
- B. Cyber Safety
- C. Data Aggregation
- D. Cuboid Lattice

Answer: B. Cyber Safety

18. In web mining, the process of merging image content with surrounding text descriptions to find hidden relationships is an example of:

- A. Data Pre-processing
- B. Data Fusion
- C. Data Cube Materialization
- D. Normalization

Answer: B. Data Fusion

19. "The 'Curse of Dimensionality' refers to how adding more dimensions makes distance-based separation more effective in high-dimensional space." — Is this statement true or false?

- A. True
- B. False

Answer: B. False (It makes separation "fuzzy" because points appear equally far away)

20. Which graph mining task involves predicting whether a new connection will form between two existing nodes in a social network?

- A. Community Detection
- B. Link Prediction
- C. Frequent Subgraph Mining
- D. PageRank

Answer: B. Link Prediction